CONTINUED

daily processes in these organisations. One example of this was the National Health Service (NHS) in Britain. British hospitals were forced to turn away some patients who could not be treated until their personal information could be accessed again.

WannaCry malware was based on a set of malware code called EternalBlue. There have been several malware attacks since WannaCry based on this same EternalBlue code.

Discussion questions

1 Do you think it is the organisation's fault that the malware infected their systems and caused issues?

2 Do you think it is completely the hacker's fault for being able to put the malware on the systems?

3 How would you feel if your files were encrypted and you couldn't access them? Would you pay the ransom, or would you take a different action?

5.1 The internet and the world wide web

The internet and the world wide web: aren't they the same thing? Although the terms are used interchangeably, they aren't the same thing. Bonus points if you knew that already! The **internet** is an **infrastructure**. This means it is all the components and cables that are used to connect one device to another. Another way that the internet can be described is that it is one huge **network**. More specifically, it is a type of network called a wide area network (WAN). This is a type of network that spans a large geographic area, in this case the whole world! The internet is an incredible structure if you think that homes and businesses all over the world are able to use it to connect to each other. Millions and millions of people use this huge infrastructure.



Figure 5.3: The internet connects millions of people

The **world wide web** is the collective term for all the **websites** and **web pages** that are available. The world wide web is accessed by using the internet.

KEY WORDS

internet: a very large global network that allows users to access the world wide web.

infrastructure: the physical structure, such as all the components and cables, that are used to connect devices.

network: computers and devices that are joined together using cables or wireless technology.

world wide web: the collection of all the web pages and websites that have been made available.

website: a collection of web pages, normally about a related topic or purpose.

web page: a single page within a website.

COMPUTER SCIENCE IN CONTEXT

The world wide web was invented by Tim Berners-Lee in 1989. He was motivated to do so by a problem that he had. He is a Computer Scientist and found that at the time he had several computers that all had different information stored on them. He found that he would have to keep logging onto the different computers to obtain the particular information that he needed. He saw a way to solve this issue that he realised could be used universally. Lots of computers were already being connected together into networks and he realised that he could share information using an emerging technology at the time called Hypertext. By October of 1990 he had developed the three fundamental parts that create the world wide web: HyperText Markup language (HTML), uniform resource locator (URL) and the Hypertext Transfer Protocol (HTTP). By the end of 1990, the first web server was set up and the first web page displaying information about the world wide web project was created.

ACTIVITY 5.1

Have a look at what the first web page looked like. Type 'the world wide web project' into your browser and look for the 'info.cern' link.



Figure 5.4: HTML

5.2 Requesting and retrieving web pages

A **URL** is a fundamental component of the world wide web. Each website and web page on the world wide web has a text based address. This address is called a URL. To access the web page, a user types the URL into the address bar of a **web browser**.

A URL has three main components: the **protocol**, the **domain name** and the web page or file name.

KEY WORDS

uniform resource locater (URL): another name for the text-based address for a website.

web browser: a piece of software that retrieves and displays web pages.

protocol: a set of rules for transmitting data.

domain name: a specific part of a URL that contains the unique name for the web server.

Figure 5.5 shows an example of a URL:

https://www.cambridge.org/education

protocol | domain name | web page name

Figure 5.5: The components of a URL

There is a whole process involved in requesting and retrieving web pages that you need to understand, the URL and its three components are fundamental to this process. So how is a web page requested and retrieved by our computers?

To start the process, the user opens their web browser and types the URL into the address bar. The web browser then sends the URL for the website to the **domain name server (DNS)**. The DNS is a special kind of server that stores all the different domain names (part of the URL shown in Figure 5.5) along with their equivalent IP address. The DNS looks through its database of domain names to find the equivalent IP address. As you can imagine, there are billions of websites, so these are not all stored on a single DNS. Therefore, the first DNS will search its database and if it doesn't find the domain name, it sends it to the next DNS. This will happen until the domain name is found, or if it is not found, a message is sent back to the web browser to say that the website is not found. When the DNS find the domain name, it sends the equivalent IP address back to the web browser – the IP address of the **web server** that stores the website. This process is summarised in Figure 5.6.

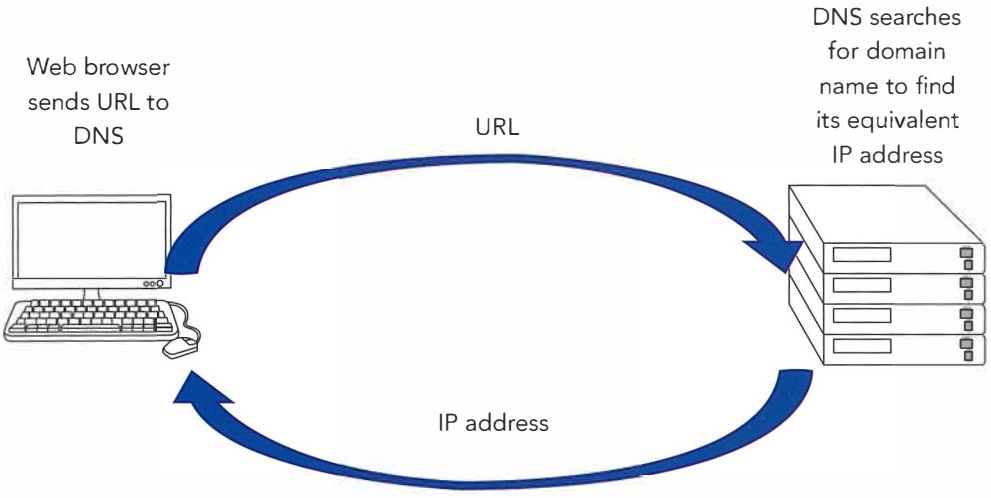


Figure 5.6: A web browser requests an IP address from a DNS

The web browser receives the IP address for the website from the DNS. It now knows where to find the website. The web browser sends a request to the web server to ask for the web page from the website. It uses a protocol called the **hypertext transfer protocol (HTTP)** to send the request to the web server. Each web page is created using **hypertext markup language (HTML)**, **cascading style sheets (CSS)** and may include **active script** such as JavaScript. When the web server receives a request for a web page, it sends the data for the website, including the HTML, CSS and any active script, back to the web browser using the HTTP. The web browser then renders the HTML and CSS and runs any active script to display the web page. This process is summarised in Figure 5.7.

KEY WORDS

domain name server (DNS): a special server that contains a database of domain names and their corresponding IP address.

web server: a network component where the web pages for a website are stored.

hypertext transfer protocol (HTTP): a protocol that is used for the transmission of web pages and related data across the internet.

hypertext markup language (HTML): a scripting (web authoring) language that is used to create web pages.

cascading style sheet (CSS): a scripting language that is used to create a presentation template for a web page. It includes what kind of font and colour text will appear on the webpage.

active script: this is the script that is used to produce any of the interactive elements of a web page.

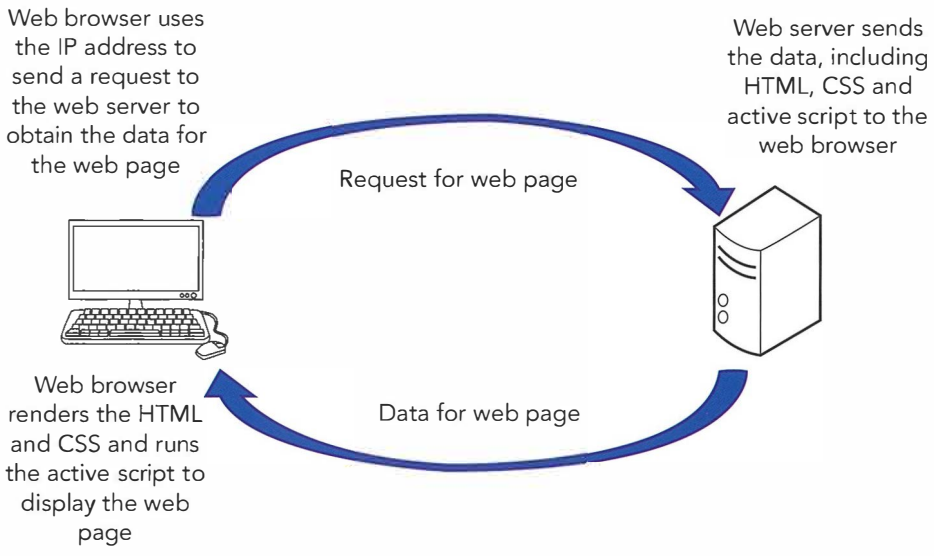


Figure 5.7: The web browser requests a web page and renders the HTML and CSS

Questions

1 What is a URL?

2 What is an IP address?

3 What are the **three** main parts of a URL?

Another protocol can be used to request the data for the web pages that will require an additional process to take place. This protocol is the **hypertext transfer protocol secure (HTTPS)**. This protocol has an additional layer of security that **encrypts** any data that is transferred. It involves the use of **digital certificates**. This is a certificate that the owners of the website need to apply for from a **certificate authority**. A certificate authority is an organisation that checks whether a website is authentic and trustworthy. If the certificate authority believe the website is authentic, they will issue them with a digital certificate. This is how the additional layer of security is created.

- Before the web browser sends a request to the web server to ask for the web pages, it sends a request to the web server asking it to provide a digital certificate.
- The web server sends a copy of the digital certificate to the web browser.
- The web browser checks if the digital certificate is authentic.
- If the certificate is authentic, the web browser will allow communication to continue with the web server and any data that is transmitted between the two is encrypted.
- If the certificate is not authentic, the web browser will report that the website is not secure.

KEY WORDS

hypertext transfer protocol secure (HTTPS): a secure version of the HTTP that encrypts data for transmission.

encryption: a method of securing data for storage or transmission that scrambles it and makes it meaningless.

digital certificate: a certificate that is awarded to a website if they can prove that they are a real organisation and take measures to securely transmit their data.

certificate authority: the awarding organisation that checks if another organisation is real and secure before awarding them a digital certificate.

This process is summarised in Figure 5.8.

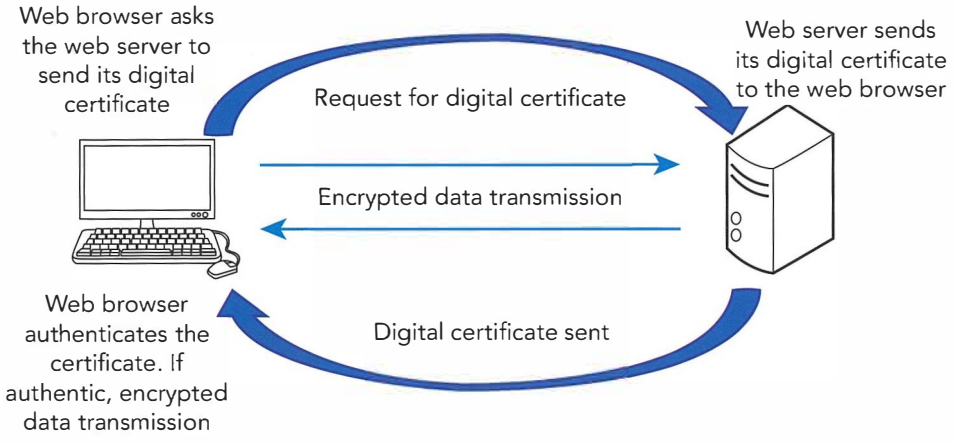


Figure 5.8: A web browser requests a digital certificate to allow encrypted data transmission

The HTTPS protocol is the standard HTTP protocol with a layer of security combined, that is either the **secure sockets layer (SSL)** or **transport layer security (TLS) protocol**. TLS is a newer version of SSL. This is the protocol that is used to create the encryption. You can check visually whether a website is secure by seeing if it uses the HTTPS protocol. If it does, you will see this at the start of the URL in the address bar in the web browser. You will also see a small padlock before the start of the URL. Most web browsers also have an option for you to view the digital certificate for the website. This is often found by right clicking your mouse on the padlock.

ACTIVITY 5.2

Use the internet to research why the TLS protocol was developed to replace the SSL protocol.

Questions

4 What is a digital certificate and how is it used in the HTTPS protocol?

5 Does the web browser or the web server authenticate the digital certificate?

KEY WORDS

secure sockets layer (SSL) protocol: a type of protocol that encrypts data for transmission. It is the protocol, that is combined with the HTTP to create HTTPS.

transport layer security (TLS) protocol: an updated version of the SSL protocol.

render: the method of processing all the web page data, such as HTML, to display the web page.

5.3 The purpose of a web browser

You may have noticed that the web browser has a key role in the requesting and retrieving of web pages. The main purpose of a web browser is that it is a software application that allows you to access information that is available on the world wide web. As you learnt previously, it does this by requesting the HTML files and other data required to create the web page, then **renders** this data so that you can view the web page.

One thing to note about a web browser is that some people confuse it with a search engine. This is because some web browser software development companies also

produce search engines too. One example of this is Google. Google produce a search engine called google, that you can use to find information on the world wide web, however this is not a browser. The browser that Google produce is Google Chrome. This is the software that allows you to open web pages, such as the google search engine. Why don't you test your friends and see if they know the difference?



Figure 5.9: Example of a search engine

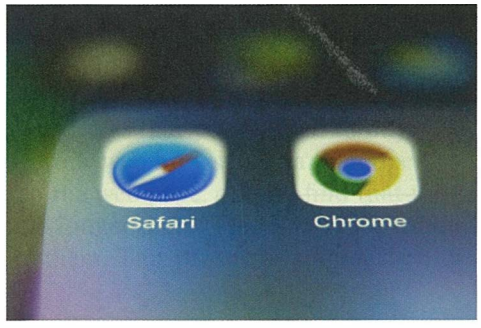


Figure 5.10: Examples of web browsers

ACTIVITY 5.3

Can you think of two more examples of a web browser? Use the internet to check if you are correct.

Did you know that there are lots of other things that a web browser can do? This is a list of some of the main functions of a web browser, how many have you used before?

- It provides an **address bar** that you can use to type in a URL.
- It keeps a record of all the web pages that you have visited. This is called **user history**.
- It allows you to **bookmark** web pages and set web pages as your favourites. These can then be displayed on the toolbar of your web browser to allow you to access them much more quickly. It means you can just click the bookmark or favourite icon and it will take you straight to the web page.
- It allows you to open multiple **tabs** at a time. The browser itself opens into a **window** on your computer. Without tabs, you would need to open multiple windows if you wanted to open more than one web page at a time. Tabs allow you to open multiple web pages in the same browser window which makes it easier to navigate between the different web pages that you have open.
- It allows you to **navigate** between web pages. The buttons that are most commonly available on a web browser are forward, back and reload the page.
- It stores **cookies** and manages the process of transmitting them. You will learn more about cookies next!

KEY WORDS

address bar: the section of a web browser where you type the web page address.

user history: a record made on your web browser of all the web pages that you have visited.

bookmark: a link that can be created to allow you to quickly find a web page again.

tab: a website function that allows you to open multiple web pages in the same window.

window: a viewing area that is created in which a software application or file can be opened.

navigate: move around web pages, or move back or forward to previous web pages.

cookie: small text file that is used to store personal data, by a web browser.

The main features are shown in Figure 5.11.

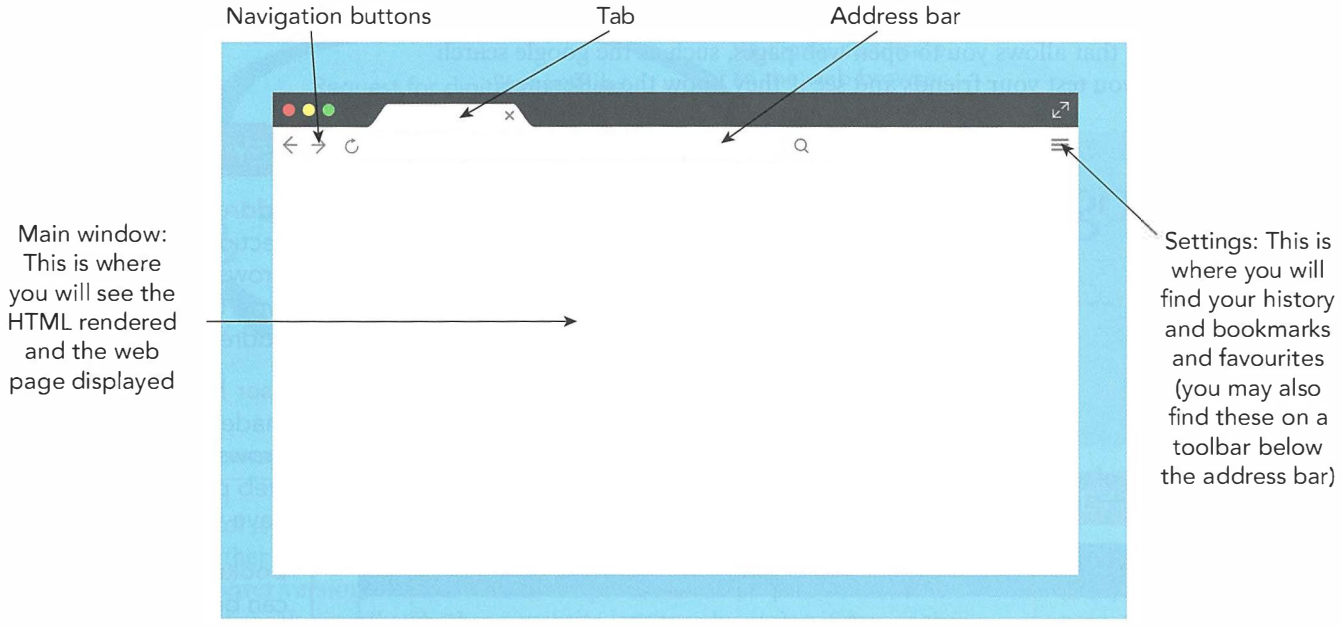


Figure 5.11: The main features of a web browser

Cookies! Sadly, not the tasty kind, but a useful function of a web browser. So, what exactly is a cookie? A very simple explanation is that it is a small text file that is used to store data, normally about you.

They are very useful as they can store data that you may regularly use on the internet. This means that you do not need to enter this data every time. Examples of this are:

- Storing your username and password for accounts such as your social media.
- Storing your bank card details for buying products and services online.
- Storing products in a virtual shopping cart when you are buying multiple products from an online company. It can also save the items in your cart so that you can buy them at a later date if you don't want to buy them now.

Cookies can be used to tailor your online experience with websites. Some websites allow you to format the website in the way that you want it to look, for example, changing the colour of text and boxes that appear on the web page. Cookies can also be used to track the different content that you look at on the world wide web. For example, if you look at web pages about your favourite hobby or interest, you will find that advertisements appear on web pages that are about your favourite hobby too. This is called targeted advertising.

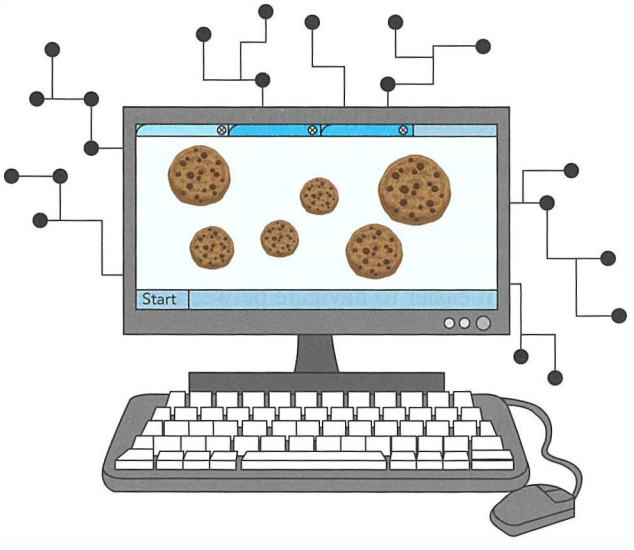


Figure 5.12: Many websites use cookies

Cookies are created when you visit a web page and then they are stored by your web browser. There are two main types of cookie, **session cookies** and **persistent cookies**.

Session cookies are temporary files that are created when you visit a web page. As soon as you close your web browser, these cookies are deleted. If you open your web browser again and go back to the same web page, the web page will not recognise you. That means that it will not be sent any cookies by your web browser, about your personal details or preferences, as this type of cookie are not saved by the browser once it is closed. You might have experienced this type of cookie when you are online shopping. You may have added several items that you wanted to buy to a shopping cart on a website, you then close your web browser. You then open your web browser at a later date and go back to the web page, only to find that the items you put in the online shopping cart are no longer there. This is because it is a session cookie that is used to temporarily store them. This process is summarised in Figure 5.13.

KEY WORDS

session cookie: a type of cookie that is deleted when the web browser is closed.

persistent cookie: a type of cookie that is stored by the web browser until it is deleted by the user or because it has expired.

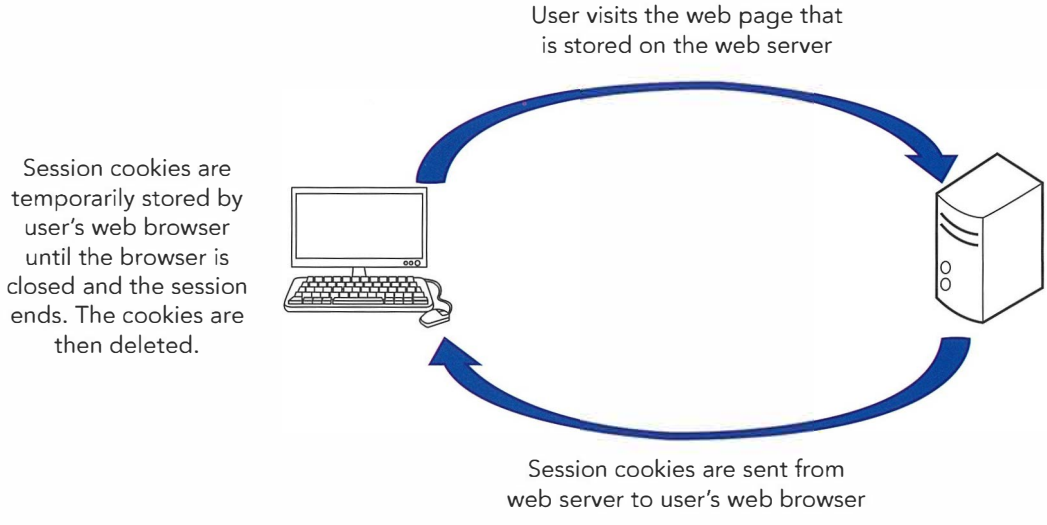


Figure 5.13: Session cookies

Persistent cookies are permanent files that are created when you visit a web page. These are stored by your web browser onto the hard drive of your computer. When you visit the same web page again, the cookie file is sent back to the web server that stores the web pages to provide the data about your personal details or preferences. This will allow your login details to be automatically entered, for example. These cookies will remain stored on your hard drive until you manually delete them, or your browser deletes them because they have expired. Each persistent cookie file is sent with an expiration date. When this expiration date is reached, the web browser will delete this cookie from your hard drive. This process is summarised in Figure 5.14.

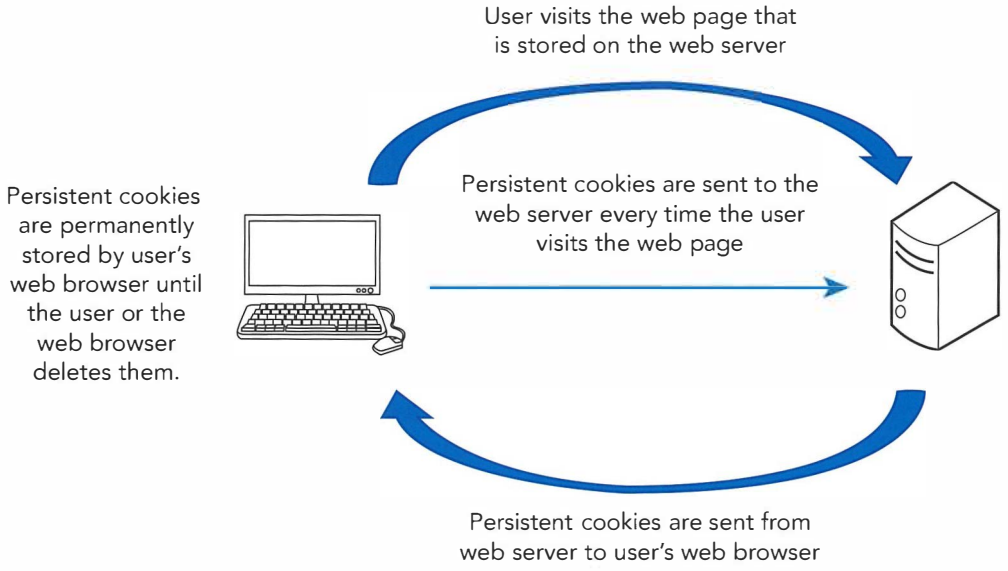


Figure 5.14: Persistent cookies

Internet users have mixed feelings about cookies. This is because they can have a very useful role but can also be used for improper activities. Cookies do build up an online profile about your details and preferences. Some people feel this is an invasion of their privacy. Other people also believe that this data can be gathered to build a profile about you and add to the risk of your identity being stolen. If you are worried about this being a possibility, you can limit the use of cookies in your online activity by changing the settings for your web browser.



Figure 5.15: Cookies can be a privacy concern

Questions

6 What is a cookie used for?

7 What is the difference between a session cookie and a persistent cookie?

8 Why do some people have an issue with cookies being used?

COMPUTER SCIENCE IN CONTEXT

Each company that creates a web browser will often have a slightly different aim as its focus. Some web browser development companies focus on the security of the transmission of data, whereas others focus on how quickly the web pages are rendered and displayed.